



Indian Institute of Information Technology (IIIT) Surat
Department of Computer Science & Engineering
B.Tech-II, Semester - III
End Semester Examination, December-2025
Subject: Database Management Systems (CS 302)

Date: 4th December, 2025

Timing: 10:00 A.M. to 01:00 P.M.

Max Marks: 100

Instructions:

1. Make suitable assumptions whenever necessary with proper justifications.
2. Figures to the right indicate maximum marks assigned for that question.
3. **Attempt all questions.**

Q.1. Construct an Entity-Relationship (ER) diagram for a "Restaurant Menu Ordering System" based on the description as follows: Customers seated at a restaurant table can browse the menu, call the waiter, place food orders, and obtain the final bill through the computer interface available at their table. Waiters, using wireless tablet devices, can initialize a table for new customers, assist them with table functions, record or modify food orders, and send these orders to the kitchen. The food preparation staff, through their touch-display terminals, can view incoming orders, update the status of each food item during preparation, and notify the waiters once items are ready to be served. Supervisors may intervene when required to manage exceptional situations such as refunds or cancelled orders.

[8]

Capture the major entities, their attributes and primary keys, and the relationships between them with appropriate min-max cardinalities in diagram. Include any weak entities or specializations/generalizations if needed. You may assume additional attributes or entities wherever logically appropriate.

Q.2. Consider the relational schema for the *Library Database* where the primary keys are underlined as given below. Give an expression in the relational algebra to express **THREE** of the following queries:

[6]

Book(book_id, title, publisher_id, pub_year)
Branch(branch_id, branch_name, branch_city)
Member(member_id, member_name, address, phone)
Author(author_id, author_name)
BookAuthor(book_id, author_id)
BookCopies(book_id, branch_id, no_of_copies)
Borrowing(trans_id, member_id, book_id, branch_id, issue_date, return_date)

- Find the titles of all books published in the year 2020.
- Find the names of members who borrowed any book from the branch "Central Library".
- Find the book titles and author names for all books written by "Chetan Bhagat".
- Find the member IDs of all members who have borrowed every book available at the "City Branch".

Q.3. Consider the above given relational schema of the *Library Database*. Give an expression in the Tuple Relational Calculus (TRC) to express **THREE** of the following queries:

[6]

- Retrieve the names of all authors who have written at least one book.
- Find the names of members who have never borrowed any book.

- Find the titles of books that are available at every branch.
- For each branch located in "Mumbai", find the names of members who borrowed books from that branch.

Q.4. Consider the above given relational schema of the *Library Database*. Write a SQL query for **SEVEN** of the following queries: [21]

- List the titles and publisher names of all books published after 2018.
- Display distinct names and phone numbers of all members who borrowed a book on '2024-01-15'.
- Find the names of authors who have written a book whose title contains the word "Life".
- List the branch name and total number of copies of the book "Database Management Systems" in all branches.
- List the members who borrowed more than 5 books in the last year.
- Find all books that have never been borrowed by any member.
- Find the top 3 most borrowed books (by count of borrowing transactions).
- Find the authors whose books are available in at least 3 different branches.
- Find the members who borrowed books authored by more than one author (multi-authored or different authors).

Q.5. Consider the below given relational schema R and list of FDs :

$R(studentID, studentName, courseID, courseName, instructorID, instructorName, grade, advisorID, advisorName, hobby)$

Functional Dependencies (FDs):

F1: $studentID \rightarrow studentName, advisorID, advisorName$

F2: $courseID \rightarrow courseName, instructorID$

F3: $instructorID \rightarrow instructorName$

F4: $studentID, courseID \rightarrow grade$

F5: $studentID \twoheadrightarrow hobby$

Using the above schema R and given functional dependencies, answer the following questions. Show all steps clearly.

- Compute the closure of the attribute $courseID^+$. [2]
- Normalize the relation R to Second Normal Form (2NF). [4]
- Starting from the 2NF decomposition, further normalize the schema to Third Normal Form (3NF). [4]
- Determine whether the obtained 3NF decomposition is in BCNF or not, and justify your answer. [2]
- Explain why the original schema R violates Fourth Normal Form (4NF). [2]

Q.6. Answer ANY THREE of the following questions. Each question carries 5 marks. [15]

- Consider relations $R(A, B)$ with 10,000 tuples, $S(B, C)$ with 20,000 tuples, block size = 1 block holds 50 tuples, and memory buffer size = 40 blocks. Compute the I/O cost for performing Block Nested-Loop Join of $R \bowtie S$.
- Consider a file consisting of 200,000 records stored in 2000 blocks must be sorted using external sorting. Given that the memory size is 20 blocks, determine the number of initial sorted runs that will be produced, the number of merge passes required to complete the sort, the total number of I/O operations involved, and the overall cost of performing the sorting process.

- (c) Derive optimized Relational Algebra expression using appropriate query rewrite rules (push selections, push projections, combine selections, reorder joins, etc.) for given SQL query:

"SELECT DISTINCT d.dept_name FROM department d, instructor i, teaches t WHERE d.dept_id = i.dept_id AND i.inst_id = t.inst_id AND t.year = 2023 AND d.building = 'NEW';".

- (d) Define functional dependency and explain the meaning of full functional dependency and partial dependency with examples.
- (e) Define embedded SQL approach. Describe how SQL statements are integrated into host programming languages such as C, C++, or Java.

Q.7. Answer ANY SIX of the following questions. Each question carries 5 marks.

[30]

- (a) Explain the ACID properties of a transaction with the help of example.
- (b) What is the difference between conflict serializability and view serializability? Provide an example where a schedule is view-serializable but not conflict-serializable.
- (c) Explain the undo/redo logging method with an example. Why is write-ahead logging (WAL) essential for recovery?
- (d) Discuss the differences between a file processing system and a database management system. Explain at least four disadvantages of the file system approach.
- (e) Describe Two-Phase Locking (2PL) protocol. How does it guarantee conflict serializability? State one limitation of 2PL.
- (f) Describe the structure of a DBMS architecture with the help of block diagram. Explain the role of the following components: (i) Query Processor, (ii) Storage Manager, and (iii) Transaction Manager.
- (g) Define following terms: (i) Primary Key (ii) Candidate key (iii) Foreign Key (iv) Weak Entity (v) Strong Entity.